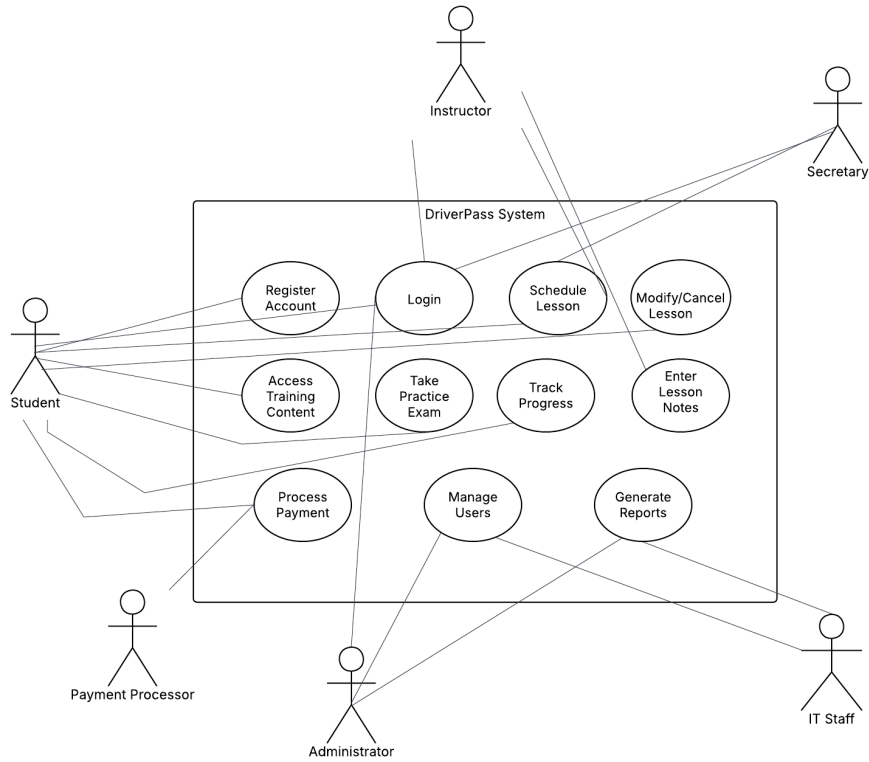


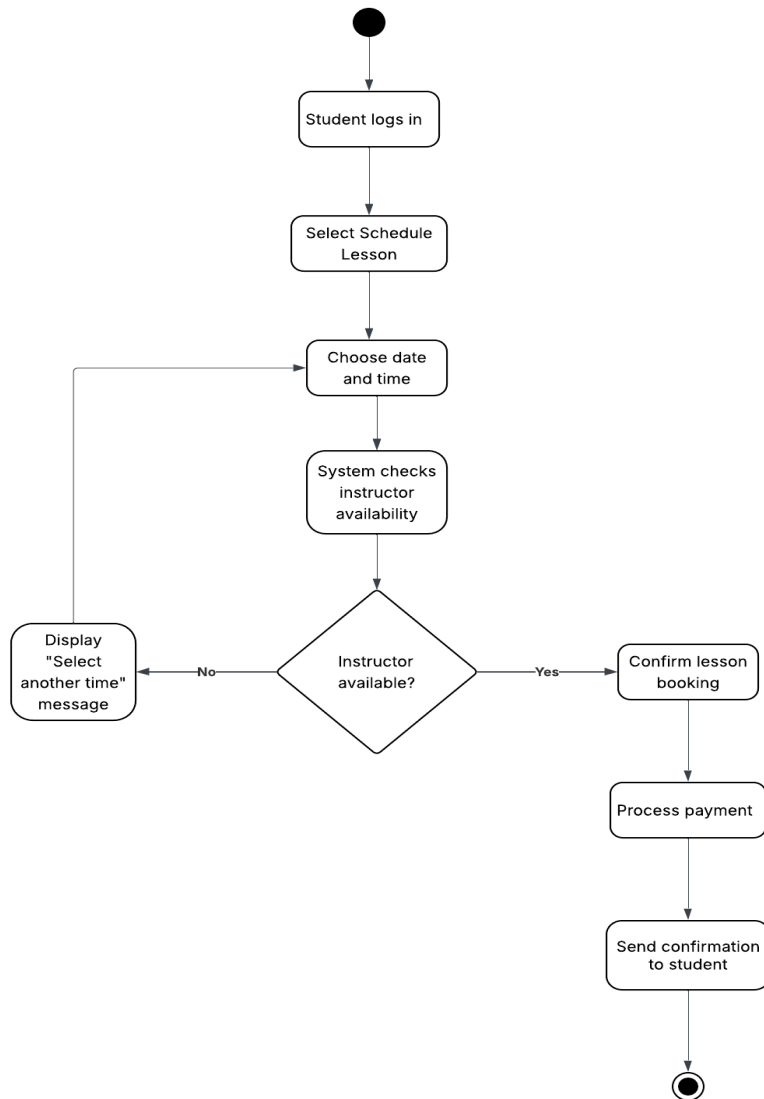
CS 255 System Design

UML Diagrams

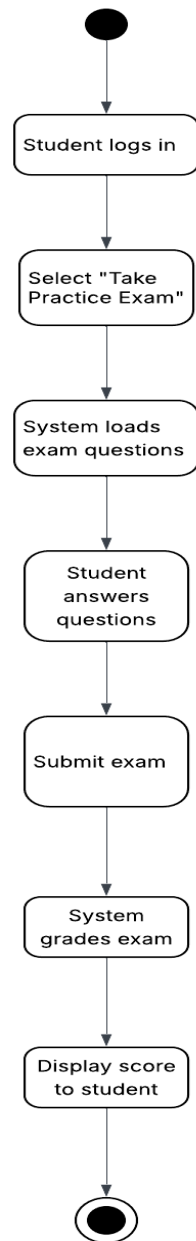
UML Use Case Diagram



UML Activity Diagram

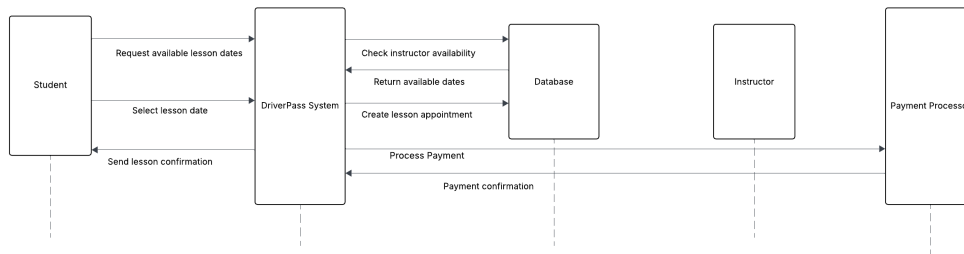


DriverPass – Activity Diagram – Take Practice Exam



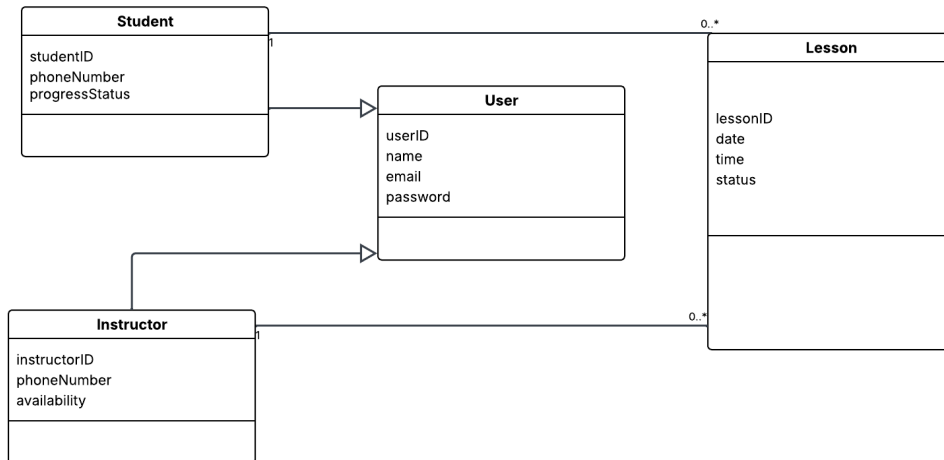
UML Sequence Diagram

DriverPass – Sequence Diagram – Schedule Lesson



UML Class Diagram

DriverPass – UML Class Diagram



Technical Requirements

The DriverPass system will require reliable hardware to support both administrators and students. Students must have access to a computer, tablet, or smartphone with internet connectivity in order to access training content, schedule lessons, and complete practice exams. Administrators and instructors will require

desktop or laptop computers to manage scheduling, monitor student progress, and generate reports. On the backend, the system will require a secure web server capable of handling user traffic and processing lesson scheduling and payment transactions.

Software Requirements

The DriverPass system will operate as a web-based application accessible through modern web browsers such as Chrome, Edge, or Safari. The system will require a relational database management system, such as MySQL or SQL Server, to store user accounts, lesson appointments, practice test results, and payment records. Secure payment gateway software will be required to process online transactions safely. Additionally, development tools such as an integrated development environment (IDE) and modeling software like Lucidchart were used to design and document the system.

Infrastructure Requirements

The DriverPass system will require secure cloud-based hosting to ensure reliability, scalability, and data protection. The application should operate over HTTPS to encrypt data transmitted between users and the system. Firewalls and intrusion detection systems should be implemented to protect sensitive student and payment information. Regular automated backups must be scheduled to prevent data loss in case of system failure. The infrastructure should also support scalability to accommodate increased user traffic as the business grows.

Security Requirements

The DriverPass system must implement secure user authentication to protect student and administrative accounts. Passwords should be encrypted and stored securely within the database. Role-based access control must be enforced so that students, instructors, and administrators can only access features relevant to their responsibilities. Payment information must be processed through a secure third-party payment processor to prevent direct storage of sensitive financial data. These measures ensure that student data, scheduling information, and

financial transactions remain protected.